Simultaneous Independent Kelly Betting with Multi-Outcome Events

Product-State Projection, Continuation Values, and Prefix Supports

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Abstract

For several independent events with finitely many outcomes, the simultaneous full Kelly problem is to maximize expected log wealth over nonnegative bets and an optional cash position. On the product state space the problem is a standard log-optimal allocation problem, but the attainable terminal-wealth profiles are not arbitrary: they are additive across events. This note isolates a simple structural consequence of that additive form. For a fixed event, every inactive outcome shares the same continuation-value factor coming from the rest of the portfolio. As a result, inactive outcomes of that event are ranked by the single-event edge ratio $p_{\ell i}/\pi_{\ell i}$, where $p_{\ell i}$ is the bettor's probability and $\pi_{\ell i}$ is the state price. We prove that, after sorting each event by this ratio, every optimal support is an eventwise prefix, and that a restricted optimum on a prefix family is globally optimal exactly when the next inactive outcome of each event fails a one-dimensional reduced-cost test. The fixed-support weights remain coupled, so unlike the one-event problem they are not generally explicit; however, a Laplace-transform representation reduces the necessary objective and gradient evaluations to one-dimensional quadrature built from eventwise factors.

Keywords. Kelly criterion; independent events; multi-outcome betting; log-optimal growth; support selection; Laplace transform.

1 Introduction

The Kelly criterion maximizes expected logarithmic growth and sits at the intersection of gambling theory, information theory, and portfolio selection [5, 2, 9, 3, 6]. In the classical horse-race or single-event multinomial problem, the optimization is explicit: after sorting outcomes by edge, one obtains a threshold rule and a closed-form stake formula [7, 8]. For several independent events played simultaneously, the problem is still concave, but the clean closed form disappears because bets on different events interact through the logarithm.

The right viewpoint is to lift the problem to the product state space. If the events are independent and event ℓ has outcome X_ℓ, then a full state is $X = (X_1, \ldots, X_m)$. In a complete market on this product space, unrestricted Kelly would choose a state-contingent wealth vector. The actual simultaneous-betting market is much more rigid: one can only buy claims tied to individual event-outcomes, so attainable wealth profiles have the additive form

$$W(x) = c + \sum_{\ell=1}^{m} g_{\ell,x_\ell}.$$

Equivalently, on the product space each bet on outcome i of event ℓ is a bundled Arrow–Debreu claim: it pays one unit in every joint state whose ℓth coordinate equals i and zero elsewhere [1, 4]. Cash is the universal claim paying one unit in every state. Thus the simultaneous Kelly problem can

be viewed as a full-bankroll allocation on synthetic instruments spanning only the additive subspace generated by these bundled claims, rather than the full space of state-contingent payoffs. The main observation of this note is that, for a fixed event ℓ, every inactive outcome of that event sees the same background wealth profile coming from the other events. That common continuation value is the multi-event analogue of cash in the single-event problem. Once it is isolated, the support structure becomes simple: within each event, positive bets can only occupy a prefix after sorting by $p_{\ell i}/\pi_{\ell i}$.

The result is not a closed-form solution of the full simultaneous problem. The fixed-support weights remain coupled across events. What the argument gives is the clean part of the structure: a short proof of the eventwise prefix property, a reduced-cost stopping rule for restricted solves, and a transform representation that makes those restricted solves computationally tractable.

2 Model and product-state viewpoint

Consider $m \geq 2$ independent events. Event $\ell \in \{1, \ldots, m\}$ has outcomes $i \in \{1, \ldots, n_\ell\}$, where $n_\ell \geq 2$. Let

$$p_{\ell i} > 0, \qquad \sum_{i=1}^{n_\ell} p_{\ell i} = 1,$$

be the bettor's subjective probabilities, and let $\pi_{\ell i} > 0$ be the state price of a unit payoff on outcome i of event ℓ. For ordinary fixed-odds betting one has $\pi_{\ell i} = 1/O_{\ell i}$, with $O_{\ell i}$ the decimal odds.

It is convenient to work with payoff units rather than stakes. If $g_{\ell i} \geq 0$ denotes the payoff contributed when event ℓ lands on outcome i, then the cost of that claim is $\pi_{\ell i} g_{\ell i}$. On the product state space, this is the payoff of the synthetic claim

$$B_{\ell i}(x) = \mathbf{1}_{\{x_\ell = i\}},$$

namely the bundle of all Arrow claims attached to joint states with $x_\ell = i$. With cash position $c \geq 0$, the budget constraint is

$$c + \sum_{\ell=1}^{m} \sum_{i=1}^{n_\ell} \pi_{\ell i} g_{\ell i} = 1. \tag{1}$$

If $X = (X_1, \ldots, X_m)$ is the realized product state, terminal wealth is

$$W(X) = c + \sum_{\ell=1}^{m} g_{\ell, X_\ell}. \tag{2}$$

Thus the simultaneous full Kelly problem is

$$\max \ \Phi(c, g) := \mathbb{E}[\log W(X)] \tag{3}$$

subject to (1), $g_{\ell i} \geq 0$, $c \geq 0$, and $W(x) > 0$ for every product state x.

Because the feasible set is compact and $\log 0 = -\infty$, an optimizer exists. The optimization is concave in the induced wealth profile. It is worth stressing that the primary object is the wealth function W, not a particular decomposition into c and g. If, for some event ℓ, the prices satisfy $\sum_i \pi_{\ell i} = 1$, then shifting a constant amount from c into every $g_{\ell i}$ leaves both budget and terminal wealth unchanged.

For later use, define the edge ratios

$$r_{\ell i} := \frac{p_{\ell i}}{\pi_{\ell i}}. \tag{4}$$

These are exactly the single-event Kelly wealth levels. In the simultaneous problem they no longer give the weights, but they still determine the support order within each event.

3 Fixed supports and continuation values

A support family is a tuple $A = (A_1, \ldots, A_m)$ with $A_\ell \subseteq \{1, \ldots, n_\ell\}$. We say that (c, g) is supported on A if $g_{\ell i} > 0$ for $i \in A_\ell$ and $g_{\ell i} = 0$ for $i \notin A_\ell$.

Fix an event ℓ. For a supported portfolio, define the background wealth from the other events by

$$R_\ell(X_{-\ell}) := c + \sum_{r \neq \ell} g_{r,X_r}. \tag{5}$$

Thus on the slice $\{X_\ell = i\}$,

$$W(X) = g_{\ell i} + R_\ell(X_{-\ell}).$$

When outcome i is inactive, $g_{\ell i} = 0$ and the slice wealth is exactly $R_\ell(X_{-\ell})$, independent of which inactive outcome is being tested.

The next proposition records the KKT system in the form needed below. Since the problem is concave, these conditions are both necessary and sufficient.

Proposition 1 (Fixed-support first-order system)**.** *Let (c, g) be optimal for the restriction of* (3) *to a fixed support family A, and let $\lambda \geq 0$ be a multiplier for the budget constraint* (1)*. Then:*

$$\mathbb{E}\left[\frac{1}{W(X)}\right] \leq \lambda, \qquad c\Big(\lambda - \mathbb{E}\left[\frac{1}{W(X)}\right]\Big) = 0, \tag{6}$$

$$p_{\ell i}\, \mathbb{E}_{-\ell}\left[\frac{1}{g_{\ell i} + R_\ell(X_{-\ell})}\right] = \lambda \pi_{\ell i} \qquad (i \in A_\ell), \tag{7}$$

$$p_{\ell i}\, K_\ell \leq \lambda \pi_{\ell i} \qquad (i \notin A_\ell), \tag{8}$$

where

$$K_\ell := \mathbb{E}_{-\ell}\left[\frac{1}{R_\ell(X_{-\ell})}\right]. \tag{9}$$

In particular, for a fixed event ℓ, all inactive outcomes share the same continuation factor K_ℓ.

Proof. The feasible set is convex and Φ is concave, so KKT suffices. Differentiate the Lagrangian

$$\mathcal{L}(c, g; \lambda) = \Phi(c, g) - \lambda \left(c + \sum_{\ell, i} \pi_{\ell i} g_{\ell i} - 1 \right).$$

For c, one obtains (6). For $g_{\ell i}$, only the slice $X_\ell = i$ contributes. By independence,

$$\frac{\partial \Phi}{\partial g_{\ell i}} = \mathbb{P}(X_\ell = i)\, \mathbb{E}_{-\ell}\left[\frac{1}{g_{\ell i} + R_\ell(X_{-\ell})}\right] = p_{\ell i}\, \mathbb{E}_{-\ell}\left[\frac{1}{g_{\ell i} + R_\ell(X_{-\ell})}\right].$$

This yields (7) for active coordinates. If $g_{\ell i} = 0$, the same derivative becomes $p_{\ell i} K_\ell$, giving the reduced-cost inequality (8). □

The proposition isolates the structural simplification bought by independence. The continuation factor K_ℓ depends on the current portfolio, but not on the particular inactive outcome i of event ℓ. That common scalar is the exact analogue of the cash cutoff in the one-event problem.

4 Prefix supports and the stopping rule

For each event ℓ, relabel outcomes so that

$$r_{\ell 1} \geq r_{\ell 2} \geq \cdots \geq r_{\ell n_\ell}. \tag{10}$$

A support family is called *prefix* if each A_ℓ has the form $\{1, \ldots, k_\ell\}$ for some $k_\ell \in \{0, 1, \ldots, n_\ell\}$.

Theorem 2 (Eventwise prefix property)**.** *Let (c, g) be an optimal portfolio for the full simultaneous Kelly problem. After sorting each event as in* (10)*, the support of (c, g) is prefix.*

Proof. Fix an event ℓ. Suppose j is active and i is inactive. From (7),

$$\lambda \pi_{\ell j} = p_{\ell j} L_{\ell j}, \qquad L_{\ell j} := \mathbb{E}_{-\ell}\left[\frac{1}{g_{\ell j} + R_\ell(X_{-\ell})}\right].$$

Since $g_{\ell j} > 0$, one has pointwise

$$\frac{1}{g_{\ell j} + R_\ell(X_{-\ell})} < \frac{1}{R_\ell(X_{-\ell})},$$

so $L_{\ell j} < K_\ell$. Therefore

$$r_{\ell j} = \frac{p_{\ell j}}{\pi_{\ell j}} = \frac{\lambda}{L_{\ell j}} > \frac{\lambda}{K_\ell}.$$

On the other hand, the inactive inequality (8) gives

$$r_{\ell i} = \frac{p_{\ell i}}{\pi_{\ell i}} \leq \frac{\lambda}{K_\ell}.$$

Hence every active outcome of event ℓ has strictly larger ratio $r_{\ell i}$ than every inactive outcome of the same event. After sorting by nonincreasing ratio, the active set must therefore be a prefix. □

For a prefix vector $k = (k_1, \ldots, k_m)$, write

$$A(k)_\ell := \{1, \ldots, k_\ell\}.$$

Assume the restricted problem on $A(k)$ has been solved, producing an optimal portfolio $(c^{(k)}, g^{(k)})$, multiplier $\lambda^{(k)}$, and continuation factors $K_\ell^{(k)}$. Since later inactive outcomes of event ℓ have smaller ratios than the first inactive one, global optimality reduces to a single test per event.

Theorem 3 (Next-outcome criterion)**.** *Fix a prefix vector k, and let $(c^{(k)}, g^{(k)})$ be optimal for the restriction to $A(k)$. Then this restricted optimum is globally optimal for the full problem if and only if*

$$p_{\ell, k_\ell+1} K_\ell^{(k)} \leq \lambda^{(k)} \pi_{\ell, k_\ell+1} \qquad \textit{for every } \ell \textit{ with } k_\ell < n_\ell. \tag{11}$$

Equivalently,

$$r_{\ell, k_\ell+1} \leq \frac{\lambda^{(k)}}{K_\ell^{(k)}} \qquad \textit{for every } \ell \textit{ with } k_\ell < n_\ell. \tag{12}$$

If (11) *fails for some event ℓ, then enlarging the allowed support by the next outcome of that event strictly enlarges the feasible region in a direction of positive reduced cost.*

Proof. If (11) holds for the first inactive outcome of each event, then by the ordering (10) it holds for every later inactive outcome of that event as well. Together with the active equalities (7) already satisfied by the restricted optimum, this is exactly the KKT system for the full problem, hence global optimality.

Conversely, if (11) fails for some ℓ, then the reduced cost of outcome $k_\ell + 1$ is positive. By concavity, moving a small amount of budget into that coordinate increases the objective, so the restricted optimum cannot be globally optimal. □

The theorem suggests a natural outer algorithm.

Corollary 4 (Monotone outer loop)**.** *Start from the empty prefix vector $k = (0, \dots, 0)$, solve the restricted problem on $A(k)$, compute $\lambda^{(k)}$ and the continuation factors $K_\ell^{(k)}$, and test* (11)*. If the test fails for one or more events, increase k_ℓ by one for any chosen subset of violated events and repeat. Because prefix lengths only increase, the procedure terminates after at most $\sum_\ell n_\ell$ expansions; when it stops, the resulting restricted optimum is globally optimal.*

The outer loop is therefore combinatorially simple. What remains nontrivial is the inner restricted solve. Unlike the one-event case, the active equations (7) are coupled across events through the random background wealths R_ℓ.

5 Laplace-transform formulas for restricted solves

The restricted problem is still a genuine nonlinear program, but independence makes its expensive expectations separable after a Laplace transform. For $t \geq 0$, define the eventwise factors

$$F_\ell(t) := \sum_{i=1}^{n_\ell} p_{\ell i} e^{-t g_{\ell i}}. \tag{13}$$

Then (2) and independence give

$$\mathbb{E}[e^{-tW(X)}] = e^{-ct} \prod_{\ell=1}^{m} F_\ell(t). \tag{14}$$

This yields one-dimensional integral formulas for the objective, the continuation factors, and the gradient.

Proposition 5 (Transform representation)**.** *For every admissible portfolio (c, g),*

$$\Phi(c, g) = \int_0^\infty \frac{e^{-t} - e^{-ct} \prod_{\ell=1}^{m} F_\ell(t)}{t}\, dt, \tag{15}$$

$$\frac{\partial \Phi}{\partial g_{\ell i}} = p_{\ell i} \int_0^\infty e^{-t(c+g_{\ell i})} \prod_{r \neq \ell} F_r(t)\, dt, \tag{16}$$

$$K_\ell = \int_0^\infty e^{-ct} \prod_{r \neq \ell} F_r(t)\, dt, \tag{17}$$

$$\mathbb{E}\left[\frac{1}{W(X)}\right] = \int_0^\infty e^{-ct} \prod_{\ell=1}^{m} F_\ell(t)\, dt. \tag{18}$$

Proof. The factorization (14) is immediate from independence. The Frullani representation

$$\log u = \int_0^\infty \frac{e^{-t} - e^{-ut}}{t}\, dt \qquad (u > 0)$$

gives (15) after taking expectations and using (14). The identities

$$\frac{1}{u} = \int_0^\infty e^{-ut}\, dt \qquad (u > 0)$$

then yield (16), (17), and (18) by substitution of (14) and differentiation under the integral sign. □

The computational significance is immediate. A naive evaluation of the objective or gradient sums over the full product state space. The transform formulas replace that exponential sum by one-dimensional quadrature built from the eventwise factors F_ℓ. Once the values $F_\ell(t_j)$ are known on a quadrature grid $\{t_j\}$, all continuation factors K_ℓ can be assembled with prefix-suffix products across events.

6 Discussion

The single-event multinomial Kelly problem is explicit because inactive states all share the same cash floor, so active bets simply top up favorable outcomes to their Kelly wealth levels. The simultaneous independent-event problem is not explicit in that sense. Even on a fixed support, the optimal weights are coupled across events and must be solved numerically.

What survives, and what this note isolates, is the support logic. On the product state space, the unattainable benchmark is an arbitrary state-contingent wealth vector; the attainable family is additive across events, equivalently the span of cash and bundled Arrow claims attached to individual event-outcomes. That additive structure forces every inactive outcome of a fixed event to share the same continuation factor, and that one observation is enough to recover the eventwise prefix theorem and the reduced-cost stopping rule. In short: the outer combinatorics are simple because independence compresses the support test for each event to a single scalar threshold.

This viewpoint also suggests the right division of labor computationally. Support discovery is handled by the prefix theorem and the next-outcome test; the inner restricted solve is handled by the transform representation. The first is essentially a structural theorem. The second is a numerical convenience. Together they turn the simultaneous problem from an exponential state-space optimization into a sequence of restricted concave programs whose expensive expectations can be evaluated from one-dimensional quadrature.

References

[1] K. J. Arrow, Le rôle des valeurs boursières pour la répartition la meilleure des risques, *Econométrie* (1953), 41–47.

[2] L. Breiman, Optimal gambling systems for favorable games, in *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability, Vol. I*, University of California Press, Berkeley, 1961, pp. 65–78.

[3] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley-Interscience, Hoboken, NJ, 2006.

[4] G. Debreu, *Theory of Value: An Axiomatic Analysis of Economic Equilibrium*, Wiley, New York, 1959.

[5] J. L. Kelly, Jr., A new interpretation of information rate, *Bell System Technical Journal* **35** (1956), 917–926. doi:10.1002/j.1538-7305.1956.tb03809.x.

[6] L. C. MacLean, E. O. Thorp, and W. T. Ziemba (eds.), *The Kelly Capital Growth Investment Criterion: Theory and Practice*, World Scientific, Singapore, 2011.

[7] B. Rosner, Optimal allocation of resources in a pari-mutuel setting, *Management Science* **21** (1975), no. 9, 997–1006. doi:10.1287/mnsc.21.9.997.

[8] P. B. Smoczynski and D. Tomkins, An explicit solution to the problem of optimizing the allocations of a bettor's wealth when wagering on horse races, *The Mathematical Scientist* **35** (2010), no. 1, 10–17.

[9] E. O. Thorp, Portfolio choice and the Kelly criterion, in *Proceedings of the Business and Economics Statistics Section*, American Statistical Association, 1971, pp. 215–224.